LM5176 `EN/UVLO` Threshold with TLV431 Latch: Equations, Ranges, and Design Constraints

USB-C PD → 12 V Fan-Array Controller — Buck-Boost 42 W

Abstract

This document derives the equations and worst-case design constraints for the three-resistor `EN/UVLO` divider that feeds both the LM5176 enable pin and the TLV431 reference of the external lockout latch. It is the analytical companion to `10a_uvlo_resistor_search.py`, which searches stocked resistor values against these constraints. The latch circuit itself is documented in `uvlo_latch.tex` / `10b_uvlo_latch.py`.

Result: $R_2/R_{1a}/R_{1b} = 420/150/15\,\mathrm{k\Omega}$ ($R_2 = 120\,\mathrm{k} + 300\,\mathrm{k}$ in series), guaranteed turn-on band 2.75–4.49 V, per-unit gaps 52/41 mV.

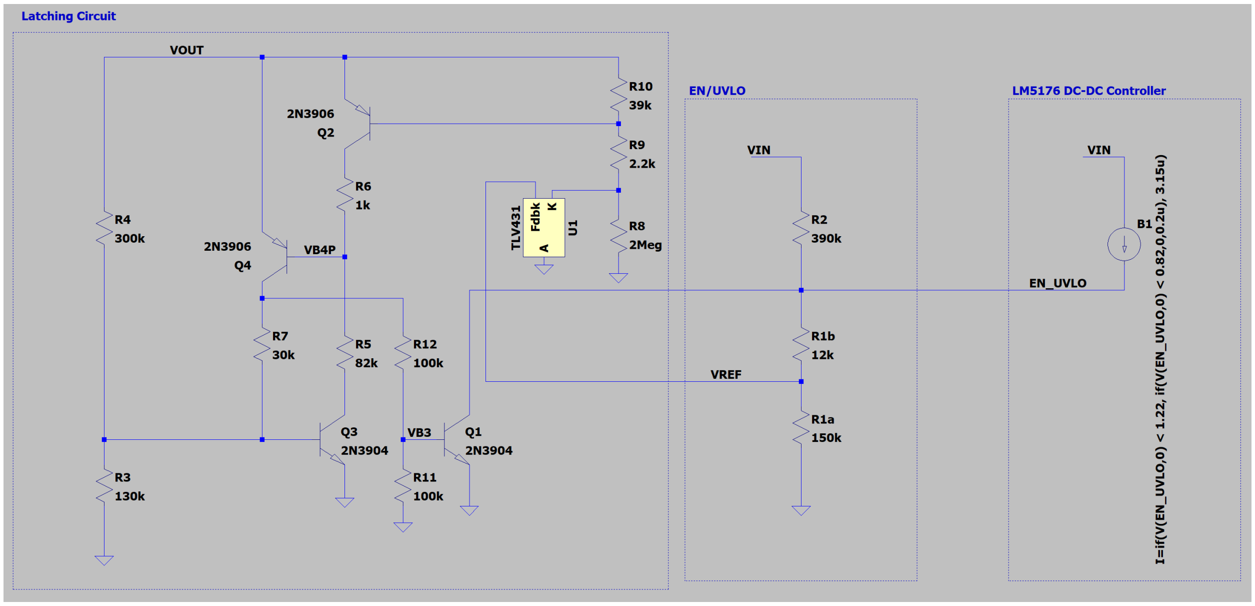
1 Motivation: why an external lockout latch

The LM5176 advertises pre-biased startup, but in this design it does not survive an input dropout in practice. The control loop is deliberately slow (low crossover frequency, chosen so the loop does not react to the fan commutation pulses — see the stability notebook), and with that loop the part misbehaves when V_{IN} recovers while the output capacitors are still charged: the restart into a pre-biased output is not handled cleanly. TI support was unable to root-cause the failure; their only suggestion was to raise the crossover frequency, which is ruled out by the system requirement that set the low crossover in the first place — the loop must not chase the fan commutation pulses, so raising its bandwidth would trade one problem for another.

The chosen fix is a small discrete latch: on an input dropout it pulls `EN/UVLO` low and, because it is *powered from* V_{OUT}, it holds `EN/UVLO` low until the output has actually discharged. Every restart therefore begins from a nearly dead output and the pre-bias path is never exercised. This document sizes the divider that tells the latch *when* to fire; the latch itself is in `uvlo_latch.tex`.

2 Circuit topology and conventions

A single node, `EN/UVLO` (voltage V_{EN}), is driven from V_{IN} through R_2. A two–element lower leg returns to ground: R_{1b} connects `EN/UVLO` to the TLV431 reference node (voltage V_{REF}), and R_{1a} connects that reference node to ground. The TLV431 `REF` pin sits on the V_{REF} node and draws a bias current I_{ref} *into* the pin.



(Left block: the lockout latch of `uvlo_latch.tex`; right: the $R_2/R_{1b}/R_{1a}$ divider on the `EN/UVLO` pin with the behavioural EN-current source used in simulation.)

Current–source convention (LM5176). Per the datasheet, the LM5176 *sources* current *out of* the `EN/UVLO` pin into the node: I_{STBY} in standby, plus an additional I_{HYS} once the operating (UVLO) threshold is crossed, which lifts V_{EN} and creates the turn-on/turn-off hysteresis. The total injected current is

$$I_S \;=\; I_{\mathrm{STBY}} + I_H, \qquad I_H = \begin{cases} 0 & \text{standby (rising, before turn-on)} \\ I_{\mathrm{HYS}} & \text{running (after turn-on)} \end{cases}$$

3 Parameter ranges

Symbol	Min	Max	Notes
V_{IN}	4.5 V	22 V	USB input
$V_{\mathrm{EN,th}}$ (operating)	1.17 V	1.29 V	LM5176 `EN/UVLO` operating threshold
I_{STBY}	1 µA	3 µA	standby source current
I_{HYS}	2.15 µA	4.25 µA	added after turn-on
I_{ref}	0 µA	0.38 µA	TLV431 `REF` input current
V_{TLV}	1.193 V	1.266 V	effective TLV431 reference (see below)
Resistors	±3%		budget-derived; see below

Effective TLV431 reference V_{TLV}. Built up from the nominal 1.24 V:

$$V_{\mathrm{TLV}}^{\max} = 1.240 + \underbrace{0.0062}_{+0.5\%} + \underbrace{0.020}_{\text{temp}} = 1.266\,\mathrm{V}, \qquad V_{\mathrm{TLV}}^{\min} = 1.240 - \underbrace{0.0062}_{-0.5\%} - \underbrace{0.020}_{\text{temp}} - \underbrace{0.021}_{V_{KA}\text{ effect}} = 1.193\,\mathrm{V}.$$

The V_{KA} term uses the datasheet reference shift $\approx -1.5\,\mathrm{mV/V}$ across the cathode excursion. The cathode floats to $\approx 15.4\,\mathrm{V}$ once the latch fires (see `uvlo_latch.tex`), so the shift spans 1.24 $\rightarrow$

15.4 V: 1.5 mV/V × 14.2 V ≈ 21 mV. The term only shifts the reference *down*, so it appears in the min only.
This excursion also drove the choice of vendor: the reference sees the full ≈ 15.4 V on its cathode whenever the latch is engaged, so a part rated to only 6 V on V_{KA} (the TI TLV431) would be outside its absolute maximum in normal operation of this circuit. The onsemi TLV431, rated 16 V, is the part assumed throughout — the two are otherwise pin- and function-compatible, which makes this an easy substitution to get wrong.

Resistor tolerance budget. ±3% everywhere: 1% purchase tolerance + ∼0.4% tempco (100 ppm/°C over the 0–50°C range) + ∼1.5% load-life and soldering drift.

4 Node equations (KCL)

`EN/UVLO` node.

$$\frac{V_{IN} - V_{\mathrm{EN}}}{R_2} + I_S \;=\; \frac{V_{\mathrm{EN}} - V_{\mathrm{REF}}}{R_{1b}} \tag{1}$$

`REF` node.

$$\frac{V_{\mathrm{EN}} - V_{\mathrm{REF}}}{R_{1b}} \;=\; \frac{V_{\mathrm{REF}}}{R_{1a}} + I_{\mathrm{ref}} \tag{2}$$

5 Closed-form solutions

Solving (1)–(2):

$$\boxed{V_{\mathrm{EN}} \;=\; \frac{V_{IN}(R_{1a}+R_{1b}) + I_S\, R_2(R_{1a}+R_{1b}) - I_{\mathrm{ref}}\, R_{1a}\, R_2}{R_{1a}+R_{1b}+R_2}} \tag{3}$$

$$\boxed{V_{\mathrm{REF}} \;=\; \frac{R_{1a}\,(V_{\mathrm{EN}} - I_{\mathrm{ref}}\, R_{1b})}{R_{1a}+R_{1b}}} \tag{4}$$

TLV431 trip relation (the key design lever). The map between V_{EN} and V_{REF} depends only on the lower leg, *independent of V_{IN}, R_2, and the EN-pin currents*:

$$V_{\mathrm{EN}} \;=\; V_{\mathrm{REF}}\,\frac{R_{1a}+R_{1b}}{R_{1a}} + I_{\mathrm{ref}}\, R_{1b} \quad\Longrightarrow\quad \boxed{V_{\mathrm{EN}}\big|_{\mathrm{trip}} \;=\; V_{\mathrm{TLV}}\,\frac{R_{1a}+R_{1b}}{R_{1a}} + I_{\mathrm{ref}}\, R_{1b}} \tag{5}$$

at the instant $V_{\mathrm{REF}} = V_{\mathrm{TLV}}$.

Inverting (3) for V_{IN}. For any target V_{EN} in a given state,

$$V_{IN} \;=\; \frac{V_{\mathrm{EN}}(R_{1a}+R_{1b}+R_2) - I_S\, R_2(R_{1a}+R_{1b}) + I_{\mathrm{ref}}\, R_{1a}\, R_2}{R_{1a}+R_{1b}}. \tag{6}$$

6 Latch behavior and the three threshold events

The external latch is powered from V_{OUT} and defaults to engaged: it is held off (*dearmed*) only while $V_{\mathrm{REF}} > V_{\mathrm{TLV}}$. There is no "arm-after-first-release" grace — whenever V_{REF} sits below V_{TLV} with the converter running *and V_{OUT} above the latch's arm threshold (≈ 2.8 V nominal; see `uvlo_latch.tex`)*, the latch grabs and pulls `EN/UVLO` low, holding until V_{OUT} has decayed (≈1–1.8 V release). The divider must therefore keep the TLV431 conducting throughout normal operation.

1. V_{IN} rises; V_{EN} (standby, $I_S = I_{\mathrm{STBY}}$) reaches $V_{\mathrm{EN,th}}$: LM5176 turns on.
2. I_{HYS} switches in $\Rightarrow$ V_{EN}, hence V_{REF}, jumps up. Because the design forces $V_{\mathrm{latch}} < V_{\mathrm{on}}$ (constraint C2), V_{REF} is already above V_{TLV} at this instant: the latch is dearmed *at* turn-on, even on an arbitrarily slow rise.
3. V_{IN} falls. While running, V_{REF} falls; when it drops below V_{TLV} the latch engages before the LM5176 would reach $V_{\mathrm{EN,th}}$ on its own (constraint C3).

Event definitions. All three events are **values of V_{IN}**, each defined by (6) with the V_{EN} level and EN-pin current of its own operating state:

Event	V_{IN} at which…	I_S	state
V_{on}	$V_{\mathrm{EN}} = V_{\mathrm{EN,th}}$	I_{STBY}	standby, rising (turn-on)
V_{latch}	$V_{\mathrm{EN}} = V_{\mathrm{EN}}\vert_{\mathrm{trip}}$	$I_{\mathrm{STBY}} + I_{\mathrm{HYS}}$	running, falling (TLV431 trip)
V_{off}	$V_{\mathrm{EN}} = V_{\mathrm{EN,th}}$	$I_{\mathrm{STBY}} + I_{\mathrm{HYS}}$	running, falling (LM5176 self-dropout)

The required ordering, for every individual unit, is

$$V_{\mathrm{off}} \;<\; V_{\mathrm{latch}} \;<\; V_{\mathrm{on}}. \tag{7}$$

Note the chain deliberately mixes states: V_{on} uses the *standby* current because that is the state the part is actually in as V_{IN} rises; the moment the threshold is crossed, I_{HYS} engages and V_{EN} jumps *upward* (the equivalent running-state crossing is V_{off}, lower by the hysteresis), so comparing V_{latch} (a running event) against the standby V_{on} is the correct and conservative form for the turn-on race.

Equivalent chain in V_{REF}. Because running-state V_{REF} is monotonic in V_{IN}, (7) is equivalent to bracketing the TLV431 reference between the running-state V_{REF} levels at the two LM5176 events:

$$\underbrace{\frac{R_{1a}(V_{\mathrm{EN,th}} - I_{\mathrm{ref}} R_{1b})}{R_{1a} + R_{1b}}}_{V_{\mathrm{REF}}^{\mathrm{run}} \text{ at } V_{\mathrm{off}}} \;<\; V_{\mathrm{TLV}} \;<\; V_{\mathrm{REF}}^{\mathrm{run}} \text{ at } V_{\mathrm{on}}, \tag{8}$$

i.e. the TLV431 must still be conducting when the LM5176 would drop out (left), and must be conducting from the moment I_{HYS} engages at turn-on (right) — the requirement begins once I_{HYS} lifts the running-state V_{REF}, which happens as (or just before) the converter starts switching.

7 Design constraints (worst case)

Each must hold across the ±3% resistor tolerance and every chip-parameter extreme of the table above.

C1 — Guaranteed turn-on by 4.5 V. In standby,

$$\min\ V_{\mathrm{EN}}\big(V_{IN}{=}4.5,\ I_S{=}I_{\mathrm{STBY}}^{\min},\ I_{\mathrm{ref}}^{\max}\big)\ \geq\ V_{\mathrm{EN,th}}^{\max} = 1.29\,\mathrm{V}. \tag{9}$$

C2 — Latch dearmed below turn-on (per-unit).

$$\min_{\text{corners}}\big(V_{\mathrm{on}} - V_{\mathrm{latch}}\big)\ \geq\ 0. \tag{10}$$

We require $V_{\mathrm{on}} > V_{\mathrm{latch}}$ so that no band exists in which the converter runs while $V_{\mathrm{REF}} < V_{\mathrm{TLV}}$ (such a band would let the default-engaged latch grab during a slow start). Like C3, (10) is a per-unit (correlated) difference — evaluated as one subtraction per device, never as a gap between independent population bands (Section 8).

C3 — Latch fires before the LM5176's own UVLO (per-unit).

$$\min_{\text{corners}}\big(V_{\mathrm{latch}} - V_{\mathrm{off}}\big)\ \geq\ 0. \tag{11}$$

C4 — Do not turn on too early. The worst-case *earliest* turn-on must stay above $V_{\mathrm{on}}^{\min} \geq 2.5\,\mathrm{V}$, so a not-yet-settled USB rail is not loaded and the converter is not asked to boost from an impractically low input. C4 fights C2: the large R_2 that C2 demands, together with the 1 µA to 3 µA spread of I_{STBY}, smears turn-on across a $\sim$1.5 V band and drags the earliest turn-on down.

The hysteresis budget links C2 and C3. From (6), $\partial V_{IN}/\partial I_S = -R_2$ exactly, so the two $V_{\mathrm{EN}} = V_{\mathrm{EN,th}}$ crossings, whose currents differ by I_{HYS}, are separated by

$$V_{\mathrm{on}} - V_{\mathrm{off}} = I_{\mathrm{HYS}}\,R_2. \tag{12}$$

Subtracting $(V_{\mathrm{latch}} - V_{\mathrm{off}})$ from both sides,

$$\boxed{\big(V_{\mathrm{on}} - V_{\mathrm{latch}}\big)\ =\ I_{\mathrm{HYS}}R_2\ -\ \big(V_{\mathrm{latch}} - V_{\mathrm{off}}\big)} \tag{13}$$

— the two gaps share the single budget $I_{\mathrm{HYS}}R_2$. A larger R_2 widens the budget, *and* the $V_{\mathrm{EN,th}}$ ($\pm$60 mV) and V_{TLV} ($+26/-47$ mV about 1.24 V) spreads inflate the worst-case $(V_{\mathrm{latch}} - V_{\mathrm{off}})$ gap by the factor $(R_{1a} + R_{1b} + R_2)/(R_{1a} + R_{1b})$ (Section 8, Eq. (17)). Fitting that inflated gap inside the budget while leaving $(V_{\mathrm{on}} - V_{\mathrm{latch}}) \geq 0$ is what forces R_2 large — and a large R_2 means a low, soft turn-on with divider current sinking toward the EN bias currents. The search rejects the *bias-dominated* region (divider current at turn-on below the total EN bias current) because there V_{EN} becomes dominated by $I_{\mathrm{STBY}}/I_{\mathrm{HYS}}$ — currents with wide spec spreads and no drift specification — so the thresholds would smear with current spread rather than track V_{IN}. The net effect is that all four constraints together are satisfiable only in a narrow design band — a real consequence of the part tolerances, not a modelling artifact.

Worked result (±3%, corrected $V_{\mathrm{TLV}}^{\min}$). Restricting the search to JLCPCB basic-part 0603 values, *no* triple of single stocked resistors clears all four constraints. Allowing R_2 to be a series pair of two basic parts (one extra placement, still no extended-part fee) yields 19 passing triples.

Selection criterion: the per-unit gaps are guaranteed positive by construction and need no extra margin, so among passing triples the search prefers the highest worst-case *earliest turn-on* $V_{\mathrm{on}}^{\min}$

— a higher V_{on} floor eases the downstream latch's dearm-before-arm race (`SORT_MODE="von_min"`). The best triple by that criterion is

$$R_2 = 420\,\text{k}\Omega\ (120\,\text{k}\Omega + 300\,\text{k}\Omega\ \text{series}), \quad R_{1a} = 150\,\text{k}\Omega, \quad R_{1b} = 15\,\text{k}\Omega,$$

with $V_{\text{on}}^{\min} = 2.75\,\text{V}$, per-unit gaps $\min(V_{\text{latch}} - V_{\text{off}}) = 52\,\text{mV}$, $\min(V_{\text{on}} - V_{\text{latch}}) = 41\,\text{mV}$, turn-on band 2.75 V to 4.49 V (C1, C4 met).

Reading the figures. Both figures are drawn in the (S, R_2) plane with $S \equiv R_{1a} + R_{1b}$, at fixed R_{1a}; every curve is the zero-margin locus of one constraint, obtained by setting that margin to zero in (6). **C1 and C4 share a single form.** Solving (6) at $V_{\text{EN}} = V_{\text{EN,th}}$ for R_2 gives the turn-on locus

$$\boxed{R_2 = \frac{S\,\big(V - V_{\text{EN,th}}\big)}{V_{\text{EN,th}} + I_{\text{ref}} R_{1a} - I_{\text{STBY}} S}} \tag{14}$$

with $(V, I_{\text{STBY}}, V_{\text{EN,th}}) = (V_{IN,\text{on}}, I_{\text{STBY}}^{\min}, V_{\text{EN,th}}^{\max})$ for C1 and $(V_{\text{on}}^{\min}, I_{\text{STBY}}^{\max}, V_{\text{EN,th}}^{\min})$ for C4.

This is not a straight line. It passes through the origin, but the denominator *shrinks* as S grows, so the curve steepens and diverges at the vertical asymptote

$$S_\infty = \frac{V_{\text{EN,th}} + I_{\text{ref}} R_{1a}}{I_{\text{STBY}}}, \tag{15}$$

the point at which the EN bias current alone holds the node at $V_{\text{EN,th}}$ and no R_2 can delay turn-on further. For C1 ($I_{\text{STBY}}^{\min} = 1\,\mu\text{A}$) $S_\infty \approx 1.35\,\text{M}\Omega$ — far outside the plot, so C1 looks gently curved; for C4 ($I_{\text{STBY}}^{\max} = 3\,\mu\text{A}$) $S_\infty \approx 409\,\text{k}\Omega$, *inside* the four-panel window, which is the steep upward sweep at its right-hand edge. The familiar straight-line approximation $R_2 \approx S(V - V_{\text{EN,th}})/V_{\text{EN,th}}$ is only valid while $I_{\text{STBY}} S \ll V_{\text{EN,th}}$ — a 13 % error for C1 at $S = 165\,\text{k}\Omega$ and 42 % for C4, which is why the plotted curves bend.

The other two boundaries:

- **C3** $S \geq R_{1a}\big(V_{\text{EN,th}}^{\max} - I_{\text{ref}}^{\max} R_{1b}\big)/V_{\text{TLV}}^{\min}$ — a *vertical* line, independent of R_2, because the per-unit gap (17) factors into $(V_{\text{EN}}|_{\text{trip}} - V_{\text{EN,th}})$, a function of the R_{1a}/R_{1b} ratio alone, times a positive scale factor.
- **C2** $R_2 \geq \Delta S/(I_{\text{HYS}}^{\min} S - \Delta)$ with $\Delta = V_{\text{TLV}}^{\max} S/R_{1a} + I_{\text{ref}}^{\max} R_{1b} - V_{\text{EN,th}}^{\min}$ — a hyperbola with its own vertical asymptote at $S = \Delta/I_{\text{HYS}}$.

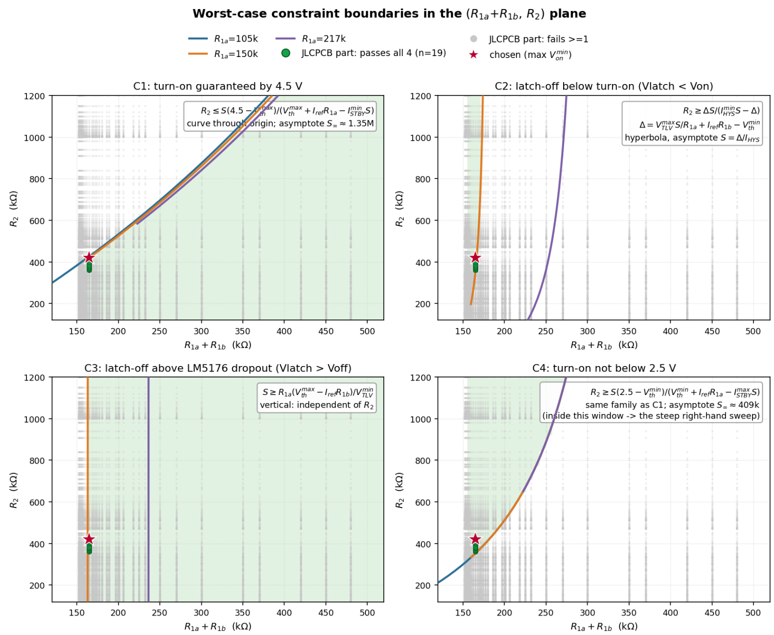
At $R_{1a} = 150\,\text{k}\Omega$, $S = 165\,\text{k}\Omega$ these evaluate to $R_2 \leq 448\,\text{k}\Omega$ (C1), $R_2 \geq 300\,\text{k}\Omega$ (C4), $R_2 \geq 298\,\text{k}\Omega$ (C2) and $S \geq 161.5\,\text{k}\Omega$ (C3) — the chosen 420/165 kΩ sits inside all four.

Equi-turn-on curves and the direction of improvement. Rearranging (6) the other way gives the earliest turn-on as an explicit function of the plot coordinates,

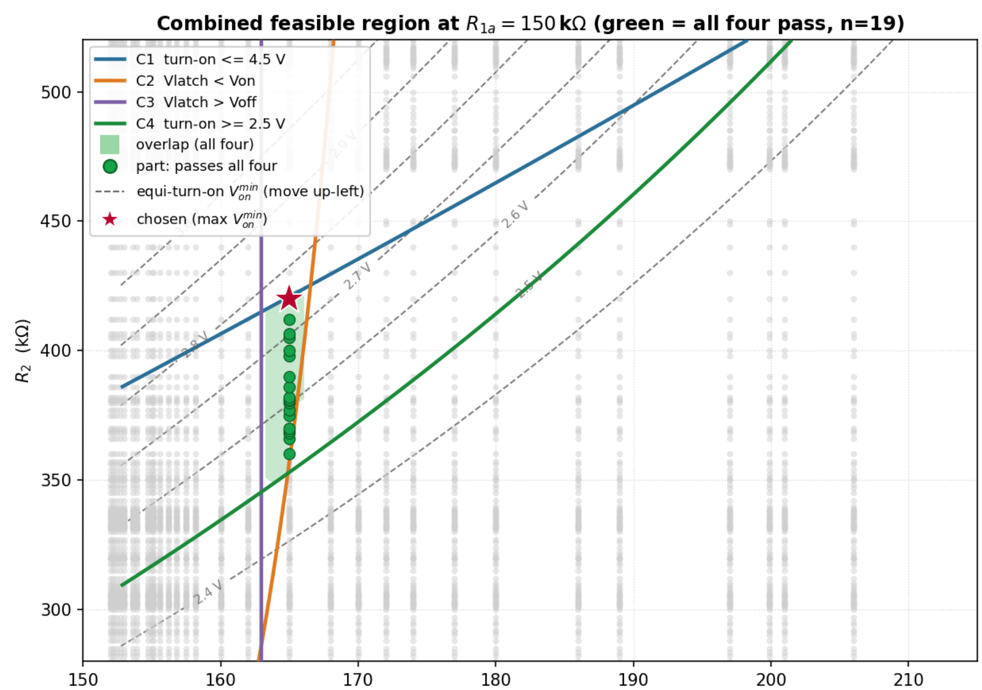
$$V_{\text{on}}^{\min} = V_{\text{EN,th}}^{\min} + \frac{R_2}{S}\Big(V_{\text{EN,th}}^{\min} + I_{\text{ref}}^{\max} R_{1a} - I_{\text{STBY}}^{\max} S\Big), \tag{16}$$

whose level sets are exactly (14) at different V — so the equi-turn-on contours are members of the *same* curved family as C1 and C4 (which are just its 4.5 V and 2.5 V members), and they share C4's 409 kΩ asymptote. The overlap figure overlays several of them (dashed grey, labelled in volts): **the design improves by moving up-left onto a higher equi-turn-on curve, until that curve leaves the shaded feasible region** — i.e. until it runs into the C1 ceiling. The chosen 420 kΩ is the highest available basic-part composite whose curve still clears C1 at the ±3% corners.

The soft, low turn-on band is the price of satisfying all four constraints at these tolerances. The star marks the chosen triple:



Shading + dots are for the middle R_{1a} (the recommended one). Dots are available JLCPCB parts; green pass all four constraints. C1/C4 (turn-on) barely move with the split; C2/C3 (latch placement) follow the R_{1a}/R_{1b} ratio and fan out. True feasible set = intersection of all four.



Boundaries are zero-margin loci. C1/C4 and the dashed equi-turn-on contours are one family, $R_2 = S(V - V_{th})/(V_{th} + I_{ref}R_{1a} - I_{STBY}S)$ -- through the origin but curving up to a vertical asymptote at $S_\infty = (V_{th} + I_{ref}R_{1a})/I_{STBY}$ ($\approx$ 409k for C4, 1.35M for C1). C3 is vertical (R_2-free); C2 is a hyperbola. Higher V_{on}^{min} = up-left, bounded by the C1 ceiling.

Cold start. C2 guarantees $V_{\mathrm{REF}} > V_{\mathrm{TLV}}$ from the instant of turn-on; the latch side of cold start (unpowered below $\approx 2\,\mathrm{V}$, dearm active before the arm threshold at every temperature) is analysed in `uvlo_latch.tex`.

8 Worst-case correlation: why the bands may overlap but units never do

It is tempting to compute the population span of each V_{IN} event independently and compare the spans. That is *too pessimistic* for the latch-vs-dropout question.

Both the LM5176 self-dropout ($V_{\mathrm{EN}} = V_{\mathrm{EN,th}}$) and the TLV431 latch ($V_{\mathrm{EN}} = V_{\mathrm{EN}}|_{\mathrm{trip}}$) are running-state events of the same device *at the same instant.* They share the identical I_{STBY}, I_{HYS}, V_{IN}, and R_2. Subtracting the two instances of (6) with a common I_S and I_{ref},

$$\boxed{V_{\mathrm{latch}} - V_{\mathrm{off}} = \left(V_{\mathrm{EN}}|_{\mathrm{trip}} - V_{\mathrm{EN,th}}\right) \frac{R_{1a} + R_{1b} + R_2}{R_{1a} + R_{1b}}} \tag{17}$$

and the bracket $V_{\mathrm{EN}}|_{\mathrm{trip}} - V_{\mathrm{EN,th}} = V_{\mathrm{TLV}}(R_{1a} + R_{1b})/R_{1a} + I_{\mathrm{ref}} R_{1b} - V_{\mathrm{EN,th}}$ contains no current term at all: I_{STBY}, I_{HYS}, V_{IN}, R_2 have all cancelled. Driving I_{HYS} to its maximum to lift V_{off} while simultaneously driving it to its minimum to drop V_{latch} describes no physical unit.

The same caution applies to C2: ($V_{\mathrm{on}} - V_{\mathrm{latch}}$) is also a per-unit difference (shared I_{ref} and resistors), and must be differenced per device, not read off the population bands. One caveat: C2's two events happen at *different times* (turn-on vs. a later dropout), so the correlation additionally assumes each unit's EN currents are stable between those moments; slow drift (e.g. temperature change between startup and the dropout) partially decorrelates C2. The C3 pair, evaluated at a single instant, has no such exposure. The accompanying script reports both the honest population bands (for visualisation) and the guaranteed per-unit gaps `gap_lo` $= \min(V_{\mathrm{latch}} - V_{\mathrm{off}})$ and `gap_hi` $= \min(V_{\mathrm{on}} - V_{\mathrm{latch}})$.

9 Simulation cross-check

`simulation/en_uvlo_lockoutv3_tlv.asc` now carries the **recommended** divider (420/150/15 kΩ) and injects the LM5176 EN-pin current via a behavioural source (0.2 µA below threshold, 3.15 µA enabled — the typical I_{HYS}). Results:

Event	Simulated	Predicted	Basis
Engage (EN clamped)	3.47 V	3.51 V	(6), $V_{\mathrm{TLV}} = 1.24$, $I_S = 3.15\,\mu$A (1.3 % error)
V_{REF} while running	1.88 V	$> V_{\mathrm{TLV}}$	TLV431 conducting (C2/C3)
Hold through V_{IN} return	EN ≤ 10 mV	latched	at $V_{OUT} = 12$ V
Release	$V_{OUT} = 1.08$ V	1.0–1.8 V	latch doc, Stage 4

The 3.47 V engage sits inside the predicted V_{latch} band and comfortably below the nominal turn-on, as C2 requires. `10b_uvlo_latch.py` re-runs these cross-checks automatically against the latest `.raw`.

10 Symbol summary

R_2	top resistor, V_{IN} →`EN/UVLO`
R_{1b}	mid resistor, `EN/UVLO`→ V_{REF}
R_{1a}	bottom resistor, $V_{\mathrm{REF}} \to$ GND
V_{EN}	`EN/UVLO` pin voltage
V_{REF}	TLV431 reference-node voltage
$V_{\mathrm{EN}}\vert_{\mathrm{trip}}$	V_{EN} when $V_{\mathrm{REF}} = V_{\mathrm{TLV}}$, Eq. (5)
$V_{\mathrm{on}}, V_{\mathrm{off}}, V_{\mathrm{latch}}$	the three V_{IN} events (table in Sec. 6)